Membrane Air Separation

Final Lab Report
Unit Operations Lab 4
November 11, 2019

Group 4
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1. Executive Summary

This lab was performed to examine the separation of O_2 gas and N_2 gas using membrane separation units that could be arranged in a series or parallel configuration. O_2 sensors and flowmeters were used with each separation unit in combination with varied system pressure to determine transport coefficients, separation factors and recoveries of each gas. In all cases O_2 recovery decreased as flowrate increased, which was the opposite of the relationship for N_2 recovery, which is expected because less separation occurs when a volume of gas is exposed to the membrane for a shorter period. O_2 and N_2 transport coefficients were generally independent of flowrate but increased with increasing pressure. Retentate based separation factors increased with increasing pressure but decreased with increasing flowrate, which is consistent with the recoveries. Additionally, series configurations yield better separation, which is expected because of longer exposure time to the membrane. The significance of these results is that predictive and meaningful separation of gases is achievable using membrane separation, thereby avoiding the need for a phase change to separate the species and reducing the cost of separation processes.

2. Introduction

The objective of this experiment is to examine the separation of O_2 gas and N_2 gas using a membrane made from hollow polymer type fibers and to determine, if possible, quantities such as: transport coefficients, separation coefficients, experimental separation factors and the recovery of each gas as a function of pressure. The significance of this experiment of this experiment comes from our ability to determine the extent to which we can separate different types of gases, and to do so while not using a phase change to achieve separation, which dramatically increases the energy costs of a process.

Our apparatus consists of a pressurized gas cylinder of dry air (at approximately 800 psig) which is regulated down to an adjustable system pressure not to exceed 150 psig. The system to which the cylinder is connected is two cylindrical shaped membrane separation units, each with a flowmeter and O_2 sensor, that can be connected either in series or parallel depending on the valve configuration. In series, each unit's shell-side O_2 level and flowrate are measured separately. In parallel, the tube side effluent streams are combined and measured (O_2 level and flowrate), as are the shell-side streams.

This type of separation process is very important in a wide variety of industrial applications such as: Hydrogen separation, separating nitrogen and air, carbon dioxide and water removal from natural gas and separation of organic vapors from air/nitrogen streams. (3) But most importantly, as stated earlier, this separation is achieved without causing a phase change (I.E. either adding the latent heat of vaporization or removing the latent heat of condensation) which dramatically increases the energy, and therefore cost, of a process.

3. Theory

Let's begin with some terminology. There are 3 relevant streams in each membrane separation unit: the feed stream, permeate stream and retentate stream. The feed stream is the influent stream which will become concentrated. The permeate stream exits the membrane unit on the opposite side of the membrane (relative to the feed) and is what the feed permeates (or diffuses) to. The retentate stream exits the membrane unit on the same side as the feed and has become concentrated because gas has diffused through the membrane. (4)

Fick's Law of diffusivity can be related to our cylindrically shaped system as follows:

$$J_A = \frac{D_A(C_{A1} - C_{A2})}{l} \quad [1] \quad (1)$$

- Where D_A is the diffusivity of species A [m^2/s]
- C_{A1} is the concentration of species A inside the membrane wall on the feed side [mol/m^3]
- C_{A2} is the concentration of species A outside the membrane wall on the permeate side [mol/m^3]
- l is the membrane thickness [m] in the radial direction
- J_A is the diffusive molar flux of species A [$mol/(m^2*s)$]

From Henry's Law, we know that the solubility coefficient is related to the concentration:

$$C_A = p_A * S_A \quad [2] \quad (1)$$

- Where p_A is the partial pressure of species A in a gas [Pa]
- S_A is the solubility constant of species A in the membrane [mol/(m3*Pa)]
- C_A is the concentration of species A inside the membrane wall [mol/m^3]

Because the penetration of a species through a membrane is a function of solubility and permeability, we define the permeability coefficient of a species as follows:

$$q_A = D_A * C_A = D_A * p_A * S_A \quad [3] \quad (1)$$

- Where q_A is the permeability coefficient of species A [mol/(m*s*Pa)]

We can then rewrite our equation for flux as:

$$J_A = \frac{D_A p_{A1} S_A - D_A p_{A2} S_A}{l} = \frac{D_A S_A}{l} * (p_{A1} - p_{A2}) = Q_A(p_{A1} - p_{A2}) \quad [4] \quad (1)$$

- Where Q_A is the Permeability of species A [$mol/(m^2*s*Pa)$]

The separation efficiency is defined as:

$$\alpha_{AB} = \frac{Q_A}{Q_B} \quad [5] \quad (1)$$

The experimental separation factor based on retentate composition is defined as:

$$\alpha'_{AB} = \frac{\frac{y_A^P}{y_B^P}}{\frac{y_A^r}{y_B^r}} \quad [6] \quad (1)$$

- Where y refers to a mole fraction
- The subscript refers to the species
- The superscript refers to the stream (Feed, Permeate or Retentate)

The experimental separation factor based on the feed composition is defined as:

$$\alpha''_{AB} = \frac{\frac{y_A^P}{y_B^P}}{\frac{y_A^f}{y_B^f}} \quad [7] \quad (1)$$

- Where y refers to a mole fraction
- The subscript refers to the species
- The superscript refers to the stream (Feed, Permeate or Retentate)

The recovery of component ‘i’ in a mixture is defined as:

$$Recovery\ of\ component\ i = \frac{\dot{V}^P * C_i^P}{\dot{V}^f * C_i^f} \quad [8] \quad (1)$$

- Where $\dot{V}$ refers to a volumetric flowrate [m^3/s]
- C refers to concentration [mol/m^3]
- The superscripts refer to the stream (feed, permeate or retentate)
- And the subscript refers to the species

The Stage Cut of a system is defined as:

$$Stage\ Cut = \frac{\dot{V}^P}{\dot{V}^r + \dot{V}^P} \quad [9] \quad (1)$$

- Where all the variables have been previously defined.

4. Results

The objective of the lab is to assess how air is affected by the membrane system. The inlet air concentration was from a compressed air cylinder containing dry air. Two different variations of the lab were run: one with the separation units in series and one in with them in parallel. The inlet pressure and flowrates were varied in both configurations. Between each trial, ample time was allowed for the oxygen levels to settle, and once settled for at least 45 seconds the data was collected. As noted in the lab manual, the feed pressure would tend to drop as the tube-side flow rate was adjusted (1). Readjustments were made to make sure the inlet pressure was always maintained at the given pressure for the trial.

Table 1 below shows the flowrates controlled for the parallel trial. The data collected for this trial can be found in appendix I in table 3, which shows the flowrates and the oxygen level from the bottom and the top.

Table 1: Parallel Trials

Parallel Configuration		
Run No.	Pressure (psig)	Flowrate (L/min)
1	20	0.5
2	20	1
3	20	2
4	20	4
5	20	8
6	40	1
7	40	2
8	40	4
9	40	8
10	80	1
11	80	2
12	80	4
13	80	8

The O_2 and N_2 recovery equations are shown in the theory section above, equation [8], with sample calculation in appendix II. These are used to calculate the recoveries shown in table 5 in appendix I. As shown in figure 1 below as pressure increases, blue line to grey line to light blue line, the amount of oxygen recovered increases. Looking at the same 3 lines, as the flow rate increased, the amount of recovered oxygen would decrease.

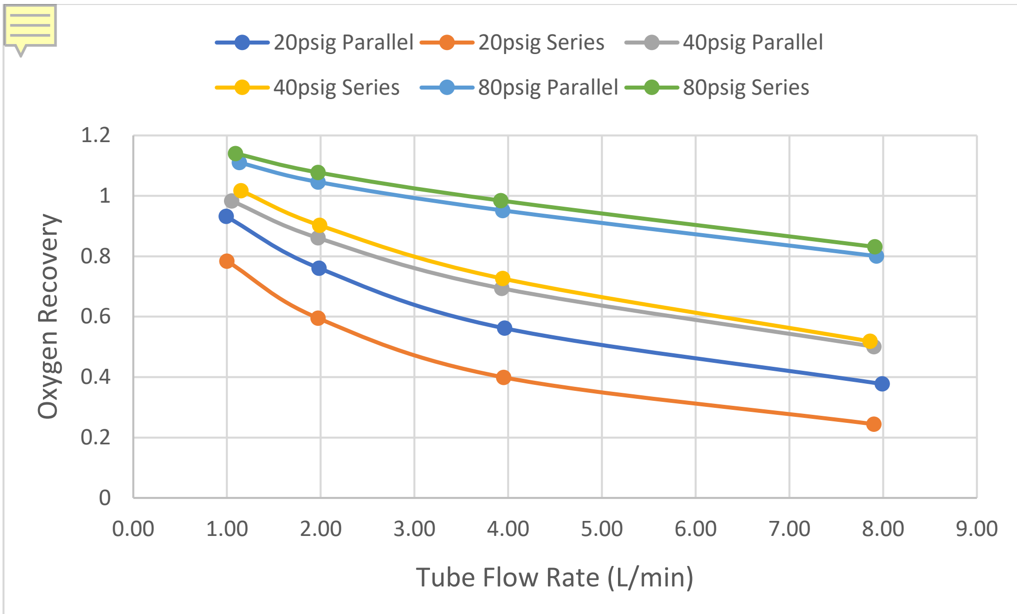

Figure 1: Oxygen Recovery vs Tube Flow Rate

Figure 2 below shows the nitrogen recovered. There is a clear relationship between the amount of oxygen and nitrogen that would come through, which is expected. As the flowrate increased the oxygen levels would decrease, however, the nitrogen levels increased. The pressures also had inverse relationship. The higher the pressure used for nitrogen the lower the recovery was.

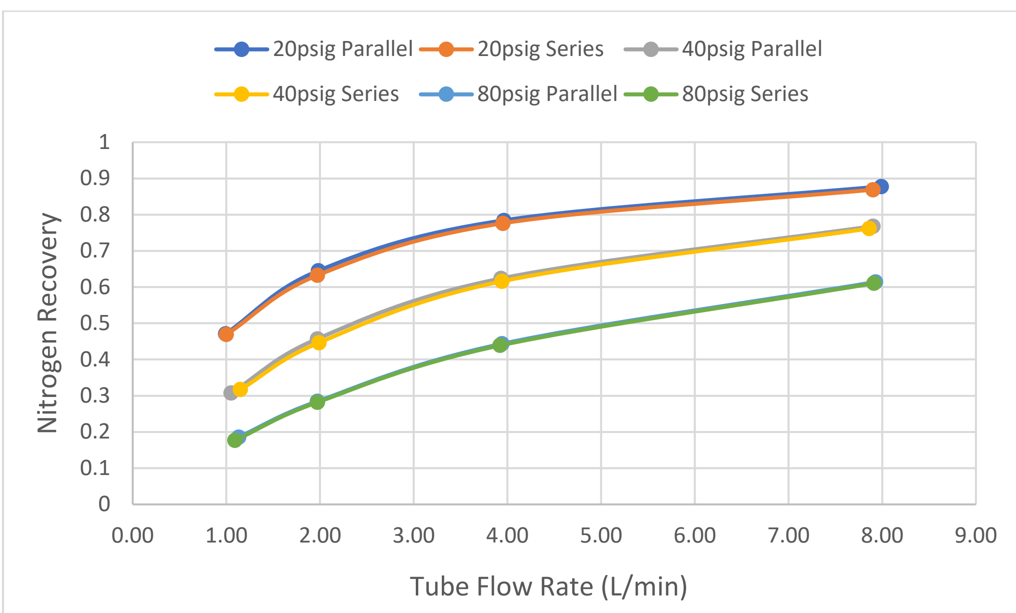

Figure 2: Nitrogen Recovery vs Tube Flow Rate

The transport coefficients (K) were also calculated and shown in table 5. This is what is used to show how much oxygen and nitrogen made it through the membrane walls.

As shown in figure 3 as the pressure increased K increased too. The other relationship seen is that as the flow rate would increase KO_2 would increase as well.

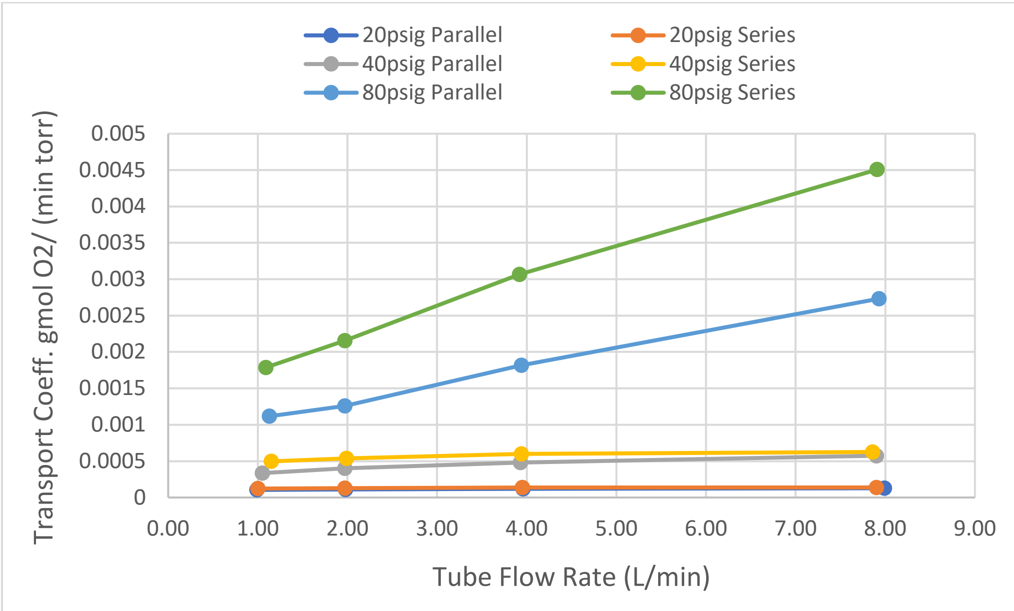

Figure 3: Oxygen Transport Coefficient vs Tube Flow Rate

Figure 4 shows that as pressure increased KN_2 also increases, which is consistent to that of what happens with KO_2. However, as the flow rate increases there is virtually no change in KN_2. To be able to see the exact relationship table 5 is used and shows that there is a slight decrease in KN_2 as the flow rate increases.

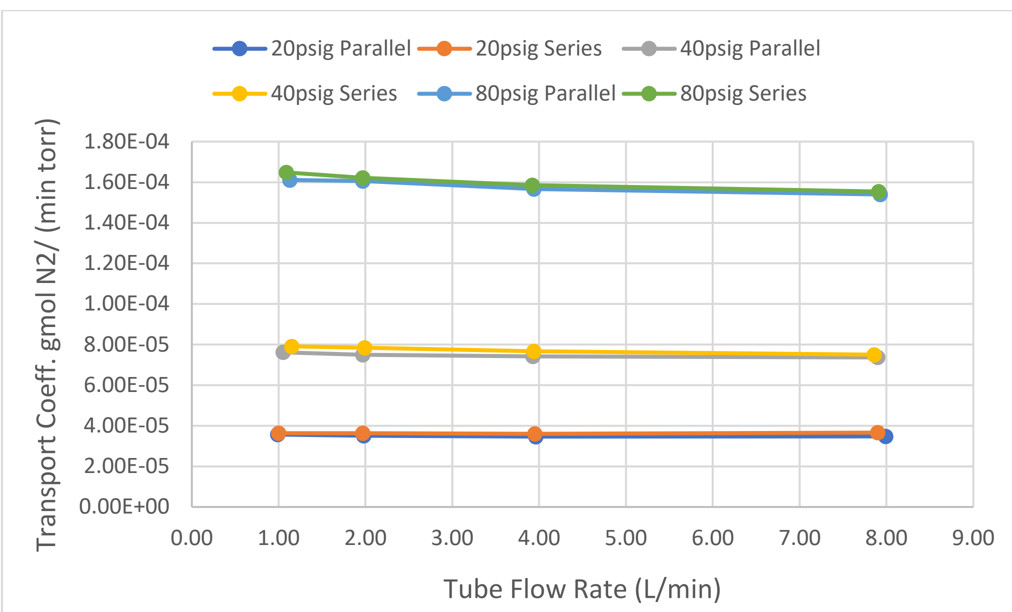

Figure 4: Nitrogen Transport Coefficient vs Tube Flow Rate

Table 2 below is the trials that were ran for the series set. The same items were collected in series as that which was collected in parallel.

Table 2: Series Trials

Series Configuration		
Run No.	Pressure (psig)	Flowrate (L/min)
1	20	0.5
2	20	1
3	20	2
4	20	4
5	20	8
6	40	1
7	40	2
8	40	4
9	40	8
10	80	1
11	80	2
12	80	4
13	80	8

The other 3 colors on the graphs: orange, yellow, and green, represent the 20, 40 and 80 psig in series respectively. Figure 1 above shows the Oxygen recovery for the series configuration. This shows the same relationship as we saw in parallel, as the pressure increases the recovery increases. As well as, as the flow rate increases the O_2 recovery decreases. In figure 2 the N_2 recovery also follows the same relationship as that in the parallel.

The transport coefficient (K) also followed the same trends in series as it did in parallel. There are two differences that are spotted in figure 3. Looking at the 40 psig trials, the series trial starts at a slightly higher KO_2 then evens out. When looking at the 80 psig there is a major difference between the series and the parallel trials. For all trials the series runs always yielded a higher KO_2. Figure 4 shows the relationship for the KN_2. Again, the same trends are found in series as they were in parallel.

Figure 5 shows the relationship between the separation factor, based off the retentate, and the flow rate. As seen in the other graphs, figure 5 shows the same trends. Parallel shows to be inferior to the series set up, leading a higher separation to be shown in the series trend. The other main relationship shown in figure 5 is that as the flow rate increases, the less separation that occurs. This does make sense in the fact that the faster you force the gas through the membrane the less time it must be sorted correctly.

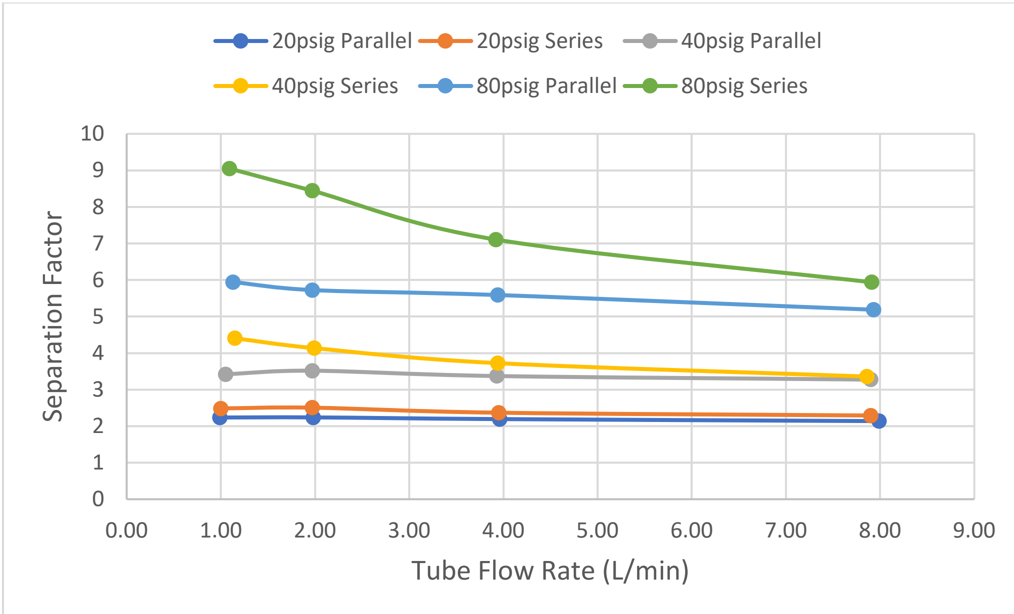

Figure 5: Separation Factor – Retentate Based vs Tube Flow Rate

Figure 6 shows the feed separation vs the flowrate. As the flowrate increases the separation factor increases. This is constant between both series and parallel. Pressure, however, does not have the same trend as it always does. In figure 6, it is seen that the more separation of the feed occurs in the 40 psig state and the 8 L/min mark. The series formation again yields higher separation in all trials.

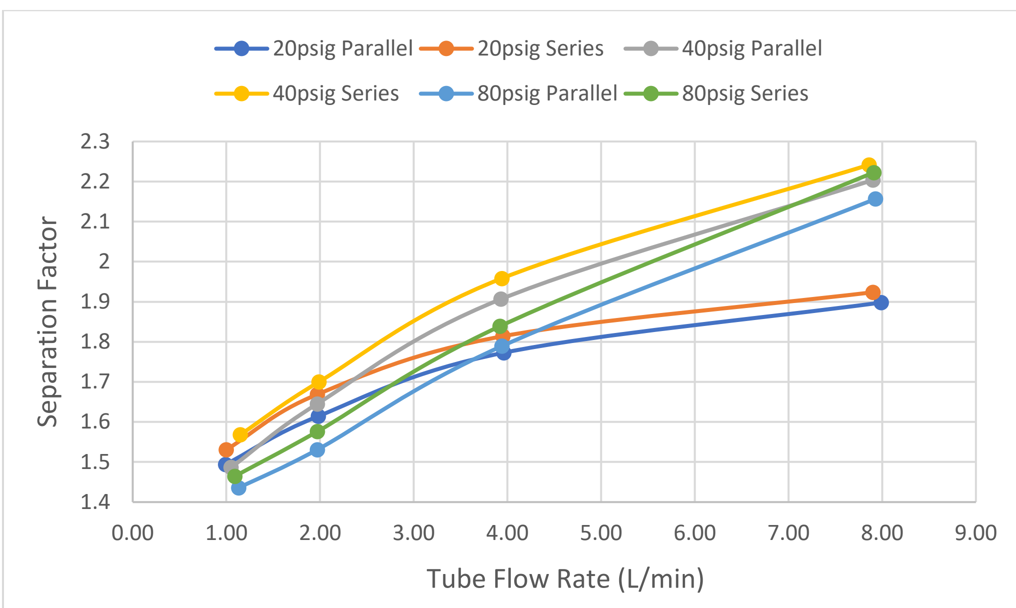

Figure 6: Separation Factor – Feed Based vs Tube Flow Rate

5. Discussion

The object of the lab was to show how change in pressure and flow rates would affect the separation of oxygen and nitrogen. Transport coefficients and recoveries for both oxygen and nitrogen were calculated. Graphs were created for these items and help prove this objective.

The separation graphs, figure 5 and 6, both yielded a higher separation for the series. Which is consistent with searched data, the lab was done for series, parallel and conventional. In which this data provided that "the conventional separation cell and the series-type separation cell yielded almost the same values, with the parallel-type separation cell falling somewhat behind." (2) Although, this lab did also run conventional, and the one that was run today did not. It stated that the conventional and the series were equal and left parallel behind, showing the relationship between series and parallel.

For figure 1 and 2, the recoveries were shown for both oxygen and nitrogen. The recoveries for oxygen in figure 1 show higher concentrations for those in series than the ones in parallel. Figure 2 was an inconclusive result since the numbers are so close together, leading the graphs to overlap.

Transport coefficients for the oxygen coefficient, figure 3, the series proved to yield a better value than that of the parallel. Again, the nitrogen graphs don't yield conclusive evidence in which would have been better.

In conclusion the system that gave the best separation is a series system with a high pressure and low feed rate. By allowing more time for the gas to sit inside the membrane more separation can occur.

6. References

(1) Unit Operations Lab: *Viscosity of Liquids*. Dr. Zdunek. Fall 2019. Department of Chemical Engineering. University of Illinois at Chicago. Lab Manual.

(2) Ohno, Masayoshi, et al. "Comparison of Gas Membrane Separation Cascades Using Conventional Separation Cell and Two-Unit Separation Cells." *Taylor & Francis.* 12 March, 2015.

(3) Freemantle, Michael. "Membranes for Gas Separation". *Chemical & Engineering News (C&EN),* vol. 83, 2005, pp. 49-57. https:/cen.acs.org/articles/83/i40/ Membranes-Gas-Separation.html.

(4) Singh, Paul. "Membrane Systems – Terminology." *YouTube*, commentary by Paul Singh, 8 Nov. 2016. https://www.youtube.com/watch?v=yc4NDVF6Q98

(5) "Standard Deviations and Normal Distribution". *Math Planet.* https://www.mathplant.com/education/algebra-2/quadratic-functions-and-inequalities/standard-deviation-and-normal-distribution.

7. Appendix I: Data Tabulation/Graphs

Table 3: Lab Data for Parallel Configuration at Various Pressures & Flowrates

Parallel Configuration					
		Flowrate (L/min)		Oxygen Level	
Run No.	Pressure (psig)	Top Flow (SHELL)	Bottom (TUBE)	Top (SHELL)	Bottom (TUBE)
1	20	1.25	0.47	26.80	14.10
2	20	1.27	0.99	28.30	15.00
3	20	1.28	1.98	29.90	16.00
4	20	1.30	3.96	31.90	17.60
5	20	1.33	7.99	33.40	19.00
6	40	2.82	1.05	28.20	10.30
7	40	2.88	1.97	30.30	11.00
8	40	3.00	3.93	33.50	13.00
9	40	3.14	7.90	36.80	15.10
10	80	6.10	1.13	27.50	6.00
11	80	6.21	1.97	28.80	6.60
12	80	6.42	3.94	32.10	7.80
13	80	6.78	7.93	36.30	9.90

Table 4: Lab Data for Series Configuration at Various Pressures & Flowrates

Series Configuration					
		Flowrate (L/min)		Oxygen Level	
Run No.	Pressure (psig)	Top Flow (SHELL)	Bottom (TUBE)	Top (SHELL)	Bottom (TUBE)
1	20	1.31	0.55	27.60	13.30
2	20	1.32	1.00	28.80	13.90
3	20	1.35	1.97	30.60	15.70
4	20	1.37	3.95	32.40	17.30
5	20	1.41	7.90	33.70	19.00
6	40	3.04	1.15	29.30	8.60
7	40	3.09	1.99	31.00	9.80
8	40	3.16	3.94	34.10	12.20
9	40	3.23	7.86	37.20	15.00
10	80	6.39	1.09	27.90	4.10
11	80	6.45	1.97	29.40	4.70
12	80	6.65	3.92	32.70	6.40
13	80	7.00	7.91	37.00	9.00

Table 5: Recoveries and Transport Coefficients for Oxygen and Nitrogen (Parallel)

Parallel Configuration						
Pressure (psig)	Tube Flow Rate (L/min)	Recoveries		Transport Coefficient		Ratio
		O2	N2	K-O2	K-N2	O2/N2
20	0.47	0.932	0.297	9.62E-05	3.59E-05	2.677
	0.99	0.761	0.471	1.05E-04	3.57E-05	2.932
	1.98	0.562	0.645	1.13E-04	3.51E-05	3.214
	3.96	0.377	0.784	1.21E-04	3.47E-05	3.497
	7.99	0.228	0.878	1.27E-04	3.48E-05	3.663
40	1.05	0.983	0.308	3.35E-04	7.62E-05	4.402
	1.97	0.861	0.457	4.01E-04	7.50E-05	5.352
	3.93	0.694	0.624	4.81E-04	7.43E-05	6.475
	7.9	0.501	0.768	5.75E-04	7.37E-05	7.801
80	1.13	1.110	0.186	1.12E-03	1.61E-04	6.931
	1.97	1.046	0.284	1.26E-03	1.61E-04	7.849
	3.94	0.952	0.443	1.82E-03	1.57E-04	11.609
	7.93	0.801	0.614	2.73E-03	1.54E-04	17.749

Table 6: Recoveries and Transport Coefficients for Oxygen and Nitrogen (Series)

Series Configuration						
Pressure (psig)	Tube Flow Rate (L/min)	Recoveries		Transport Coefficient		Ratio
		O2	N2	K-O2	K-N2	O2/N2
20	0.55	0.930	0.324	1.14E-04	3.68E-05	3.108
	1.00	0.784	0.469	1.23E-04	3.63E-05	3.377
	1.97	0.595	0.632	1.29E-04	3.64E-05	3.543
	3.95	0.399	0.776	1.36E-04	3.61E-05	3.770
	7.90	0.244	0.869	1.38E-04	3.66E-05	3.776
40	1.15	1.017	0.317	4.97E-04	7.91E-05	6.276
	1.99	0.902	0.447	5.38E-04	7.85E-05	6.859
	3.94	0.726	0.616	5.97E-04	7.67E-05	7.788
	7.86	0.518	0.762	6.26E-04	7.50E-05	8.341
80	1.09	1.140	0.177	1.79E-03	1.65E-04	10.846
	1.97	1.078	0.282	2.16E-03	1.62E-04	13.286
	3.92	0.984	0.439	3.07E-03	1.58E-04	19.357
	7.91	0.831	0.610	4.51E-03	1.55E-04	29.006

Table 7: Mole Fractions of Feed, Retentate, & Permeate at 20 psig

20 PSIG								
	Oxygen Mole Fractions			Nitrogen Mole Fractions			Separation Factor	
Configuration	x1a (Feed)	x1t (Retentate)	x1s (Permeate)	x2a (Feed)	x2t (Retentate)	x2s (Permeate)	a'	a''
Parallel	0.209	0.141	0.268	0.791	0.859	0.732	2.230	1.386
	0.209	0.15	0.283	0.791	0.85	0.717	2.237	1.494
	0.209	0.16	0.299	0.791	0.84	0.701	2.239	1.614
	0.209	0.176	0.319	0.791	0.824	0.681	2.193	1.773
	0.209	0.19	0.334	0.791	0.81	0.666	2.138	1.898
Series	0.209	0.133	0.276	0.791	0.867	0.724	2.485	1.443
	0.209	0.139	0.288	0.791	0.861	0.712	2.506	1.531
	0.209	0.157	0.306	0.791	0.843	0.694	2.367	1.669
	0.209	0.173	0.324	0.791	0.827	0.676	2.291	1.814
	0.209	0.19	0.337	0.791	0.81	0.663	2.167	1.924

Table 8: Mole Fractions of Feed, Retentate, & Permeate at 40 psig

40 PSIG								
	Oxygen Mole Fractions			Nitrogen Mole Fractions			Separation Factor	
Configuration	x1a (Feed)	x1t (Retentate)	x1s (Permeate)	x2a (Feed)	x2t (Retentate)	x2s (Permeate)	a'	a''
Parallel	0.209	0.00288	0.00139	0.791	0.897	0.718	3.420	1.486
	0.209	0.00306	0.00157	0.791	0.89	0.697	3.517	1.645
	0.209	0.00324	0.00173	0.791	0.87	0.665	3.371	1.907
	0.209	0.00337	0.0019	0.791	0.849	0.632	3.274	2.204
Series	0.209	0.086	0.293	0.791	0.914	0.707	4.404	1.568
	0.209	0.098	0.31	0.791	0.902	0.69	4.135	1.700
	0.209	0.122	0.341	0.791	0.878	0.659	3.724	1.958
	0.209	0.15	0.372	0.791	0.85	0.628	3.357	2.242

Table 9: Mole Fractions of Feed, Retentate, & Permeate at 80 psig

80 PSIG								
	Oxygen Mole Fractions			Nitrogen Mole Fractions			Separation Factor	
Configuration	x1a (Feed)	x1t (Retentate)	x1s (Permeate)	x2a (Feed)	x2t (Retentate)	x2s (Permeate)	a'	a''
Parallel	0.209	0.06	0.275	0.791	0.94	0.725	5.943	1.436
	0.209	0.066	0.288	0.791	0.934	0.712	5.724	1.531
	0.209	0.078	0.321	0.791	0.922	0.679	5.588	1.789
	0.209	0.099	0.363	0.791	0.901	0.637	5.186	2.157
Series	0.209	0.041	0.279	0.791	0.959	0.721	9.051	1.465
	0.209	0.047	0.294	0.791	0.953	0.706	8.444	1.576
	0.209	0.064	0.327	0.791	0.936	0.673	7.106	1.839
	0.209	0.09	0.37	0.791	0.91	0.63	5.938	2.223

8. Appendix II: Error Analysis

The only challenge whiling completing the lab was some of the instrumentation and devices used. Once setting a specific flow rate, the flow meter readings would fluctuate about (+/- 0.3) Std L per minute per trial. As soon as this fluctuation of the flow meter was noticed, a thirty second per trial run was created to ensure a steady state system that decreased that fluctuation to about (+/-0.1). Thus, the fluctuating of readings and sensitivity of the pressure regulator may cause a little error when performing calculations. The other device that may have caused some error in our calculation is the oxygen analyzers that would read the oxygen level going through the system. The same fluctuation as the flow meter occurred in which the same action was done to prevent less uncertainty in the readings. The following table displays the standard deviations of the separating factors and transport coefficients in the lab.

Table 10: Error Analysis

Standard Deviations		
Parameter	Parallel	Series
a'	0.154	0.664
a''	0.281	0.277
KO2	8.10E-04	1.38E-03
KN2	5.34E-05	5.37E-05

From the data shown above, most of the standard deviations were predominantly small in which indicates that the set of data points are close to the mean [5]. From some of the larger values in Table 10, that may have been due to the errors described above and assumptions made in this lab.

9. Appendix III: Sample Calculations

Tube/Shell – Side Effluent Flowrate

Example for Trial 1 at 40 psig:

$$\text{F (Tube Side)} = \text{Flowrate (Tube)} * \frac{1000}{22400} \quad (1)$$

$$\text{F (Tube Side)} = \frac{1.05\ Std\ L}{min} * \frac{1000}{22400}$$

$$\boxed{\text{F (Tube Side)} = 0.0468\ \frac{gmol}{min}}$$

$$\text{G (Shell Side)} = \text{Flowrate (Shell)} * \frac{1000}{22400}$$

$$\text{G (Shell Side)} = \frac{2.82\ Std\ L}{min} * \frac{1000}{22400}$$

$$\boxed{\text{G (Shell Side)} = 0.1259\ \frac{gmol}{min}}$$

Transport Coefficients (K) for Oxygen & Nitrogen

Example for Trial 1 at 40 psig:

$$KO_2 = \frac{g1}{p1t - p1s}$$ (1) where g1 = Permeation Rate, p1t/s = Tube/Shell Pressure

$$g1 = \frac{\text{Shell Flowrate} * \text{Mole Fraction Permeate (x1s)}}{22400} \quad (1)$$

$$x1s = \frac{Oxygen\ Level\ (Shell)}{100}$$

$$x1s = \frac{28.20}{100} = 0.282$$

$$g1 = \frac{2.82\ \frac{Std\ L}{min} * 1000 * 0.282}{22400} = 0.03555\ \frac{gmol}{min}$$

$$KO_2 = \frac{0.03555 \frac{gmol}{min}}{(320.19 - 214.32)torr}$$

$$\boxed{KO_2 = 0.0003353}$$

Same Procedure for the Nitrogen Transport Coefficient as Above

$$KN_2 = \frac{g1}{p1t - p1s}$$

$$KN_2 = \frac{0.09039 \frac{gmol}{min}}{(1732.32 - 545.68)torr}$$

$$\boxed{KN_2 = 0.00007617}$$

Recovery of Oxygen and Nitrogen

Example for Trial 1 at 40 psig:

$$O_2 \text{ Recovery} = \left(\frac{V_p C_{O_{2p}}}{V_f C_{O_{2f}}}\right) \quad (1)$$

$$O_2 \text{ Recovery} = \left(\frac{0.282 * 2.82 \frac{L}{min}}{0.209 * (2.82 + 1.05) \frac{L}{min}}\right)$$

$$\boxed{O_2 \text{ Recovery} = 0.983}$$

$$N_2 \text{ Recovery} = \left(\frac{V_r C_{N_{2r}}}{V_f C_{N_{2f}}}\right) \quad (1)$$

$$N_2 \text{ Recovery} = \left(\frac{0.897 * 1.05 \frac{L}{min}}{0.791 * (2.82 + 1.05) \frac{L}{min}}\right)$$

$$\boxed{N_2 \text{ Recovery} = 0.307}$$

<u>Separation Factors a' & a''</u>

*Subscripts: 'p' = permeate | 'r' = retentate | 'f' = feed

$$(\alpha'_{AB}) = \frac{\frac{x_{O_2p}}{x_{N_2p}}}{\frac{x_{O_2r}}{x_{N_2r}}} = \frac{\frac{0.282}{0.718}}{\frac{0.103}{0.897}} \quad (1)$$

$$\boxed{(\alpha'_{AB}) = 3.42}$$

$$(\alpha''_{AB}) = \frac{\frac{x_{O_2p}}{x_{N_2p}}}{\frac{x_{O_2f}}{x_{N_2f}}} = \frac{\frac{0.282}{0.718}}{\frac{0.209}{0.791}} \quad (1)$$

$$\boxed{(\alpha''_{AB}) = 1.48}$$

10. Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8 Hours	All
Pre-Lab Editing	1 Hours	All
Experimental Objective (Prelab)	1 Hour	Caleb
Apparatus (prelab)	2 Hours	Mike
Materials & Supplies (prelab)	1 Hour	Mike
Procedure (prelab)	2 Hours	Emir
Project Safety Evaluation (prelab)	2 Hours	Caleb
Final Lab Editing	6 Hour	All
Executive Summary (final lab)	2 Hour	Caleb
Introduction (Final lab)	2 Hours	Caleb
Theory (Final Lab)	3 Hours	Caleb
Results (final lab)	6 Hours	Mike
Discussion (final lab)	2 Hours	Mike
References (final and/or prelab)	1 Hour	All
Data Tabulation/Graphs (final)	8 Hours	Emir
Error Analysis (final lab)	2 Hours	Emir
Sample Calculations (final Lab)	5 Hours	Emir
Power Point Presentation	2 Hours	All
Total	56 Hours	